DIFFERENTIALS OVER THE BASE VANISH FOR AN IMMERSION OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1131-1135 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 1137-1142, the complete author proof (4 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The statement concerns differentials over the base S, distinct from the relative differentials of the immersion itself.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Etale-locality of the vanishing of the module of differentials (lemma-localize-differentials, same chapter) and the schemes case (Morphisms, lemma-immersion-differentials); cited inputs with the reduction written in the retained span.

Source reference. Original lemma 04CU: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 981–996. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01UY: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1142–1142. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. If $X \to Y$ is an immersion of algebraic spaces over S then $\Omega_{X/S}$ is zero.*

Proof. Follows from the schemes version, see Morphisms, Lemma 01UY, of this result via étale localization, see Lemma 04CU. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.